

Evaluating Personalized Tool-Augmented LLMs from the Perspectives of Personalization and Proactivity

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Abstract

Personalized tool utilization is essential for aligning large language models (LLMs) with user preference in interaction scenarios with various tools. However, most of the current benchmarks primarily focus on either personalization of text generation or direct tool-utilizing, without considering both. In this work, we introduce a novel benchmark **ETAPP** for evaluating personalized tool invocation, establishing a sandbox environment, and a comprehensive dataset of 800 testing cases covering diverse user profiles. To improve the accuracy of our evaluation, we propose a key-point-based LLM evaluation method, mitigating biases in the LLM-as-a-judge system by manually annotating key points for each test case and providing them to LLM as the reference. Additionally, we evaluate the excellent LLMs and provide an in-depth analysis. Furthermore, we investigate the impact of different tool-invoking strategies on LLMs’ personalization performance and the effects of fine-tuning in our task. The effectiveness of our preference-setting and key-point-based evaluation method is also validated. Our findings offer insights into improving personalized LLM agents. Our Code is available at <https://github.com/hypasd-art/ETAPP>.

1 Introduction

With the advancement of large language model (LLM) capabilities (?), more researchers are shifting their alignment objectives from targeting the general human population to focusing on specific small groups or individuals (??). Personalizing the LLMs refers to adjusting the behavior and output of LLMs to match the needs of individual users better. In this context, ?(?) introduce the concept of the *Personal LLM Agents*, which integrates personalized user data and devices to provide user with comprehensive, continuous, and personalized service. To achieve this goal, the Personal LLM

Agents must possess two key capabilities: memory (?) and interaction (??). The memory capability allows the model to track and retain both the user’s current state and their historical information, enabling more personalized and context-aware interactions. Meanwhile, the interaction capability empowers the model to engage with external systems and devices, utilizing tools to perform tasks and support the user’s needs in real-time (??).

Currently, Personal LLM Agent evaluations mainly focus on text generation (?), with limited attention to external interactions. Only a few recent studies have begun to assess Personal LLM Agents in interactive scenarios involving tool utilization. For example, ?(?) constructs a life-long personal agent framework and evaluates it using LLM. Although their work incorporates tool calling by simulating the external API execution, it does not conduct a specialized evaluation of the interaction process itself. ?(?) evaluates personalized tool usage in the specific domain of online shopping with a limited set of tools. However, these works still fail to answer the following core questions: (1) Compared to normal tool-utilizing, what evaluation metrics should be designed to assess personalized tool-utilizing? (2) How can we effectively evaluate the personalized tool usage capability in more realistic environments, beyond the narrow domain and the simulated scenarios?

To answer the first question, we argue that an ideal personal assistant should not only provide personalized services but also understand the user’s intentions and anticipate unspoken needs. It should offer comprehensive support by considering factors beyond immediate instructions, easing the user’s burden. From this perspective, Figure ?? illustrates the distinctions between traditional tool-augmented LLMs and personal tool-augmented LLMs. The latter should exhibit two key features: personalization and proactivity. **Personalization** ensures the model tailors its responses and tool usage based

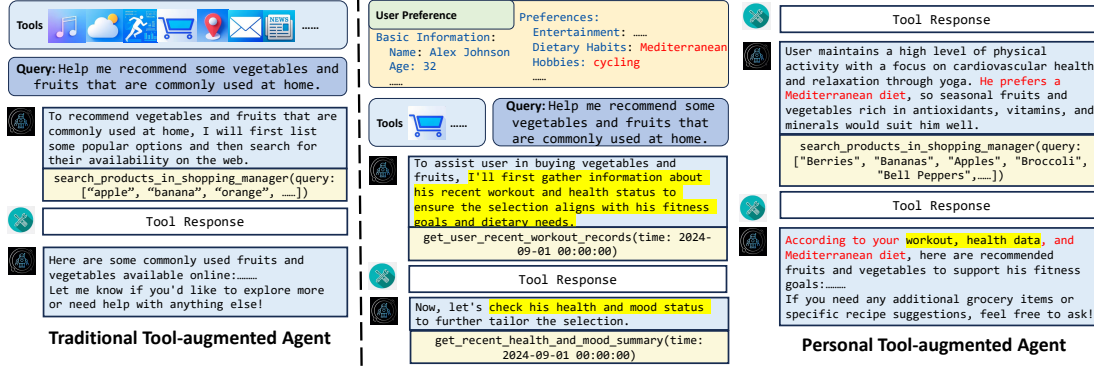


Figure 1: The difference between a traditional tool-augmented agent and a personal tool-augmented agent. Red font represents output reflecting personalization, while yellow background font represents output reflecting proactivity.

on the user’s preferences and needs. For example, when the user asks for food recommendations, the personalized LLM considers that the user prefers a Mediterranean diet (highlighted in red font), in contrast to traditional agents that offer popular options. **Proactivity** refers to the model anticipating and suggesting actions beyond the user’s request to help complete tasks more comprehensively. For instance, the assistant checks the user’s recent health status and workout records to further tailor the recommendation (highlighted in yellow background), unlike traditional models, which overlook these factors. This action is not required by the user but effectively enhances the quality of services.

Based on the two metrics mentioned above, we construct a new benchmark called **Evaluation of Tool-augmented Agent from the Personalization and Proactivity Perspective (ETAPP)** to evaluate the personalized tool invocation capabilities of large language models. For the tool-invoking framework, we build a simple sandbox to ensure the stability of the testing process, comprising the following key elements: (1) **Environment Setup:** We develop a tool-invoking system with 33 functional APIs (e.g., *add_calendar*, *view_calendar*, *get_weather*) belonging to 9 categories (e.g., *Calendar*, *Weather*), including software APIs and hardware APIs. To enhance the authenticity and stability of the evaluation, we design a simple sandbox environment where the API responses are not influenced by external environmental changes. (2) **Memory Building:** The memory of personal LLM agents includes long-term user preferences and short-term user status. We divide user preferences into two categories: high-level user profiles and low-level tool-utilizing preferences, which help us

to capture user needs more precisely. We generate 16 different user profiles based on professional backgrounds, which serve as the foundation for generating diverse evaluation samples. To simulate real usage scenarios, we construct 9 days of interaction history for each necessary tool and user. (3) **Instruction Construction:** We manually label 50 test instructions, and combine them with 16 different user profiles to create a final dataset of 800 testing cases. This data supports our evaluation of personalized tool invocation capabilities from two perspectives: personalization and proactivity.

To address the second question, we use LLM as the evaluating model to score the performance of the tool-invoking system (??). To improve the evaluation reliability, we design a key-point-based LLM evaluation method. Each testing instruction is associated with several key points annotated by humans, which serve as standard indicators for task completion. These key points are provided to the evaluation LLM to assist in scoring. The evaluation model first analyzes whether each point is satisfied and provides final analysis and score. Experimental results demonstrate that these key points significantly enhance the evaluation accuracy.

Finally, we evaluate current excellent LLMs with tool invocation capabilities on the entire set and some reasoning models (e.g., o1-mini, DeepSeek-R1) on a testing subset. The results may suggest that LLMs tend to answer the question directly without deeply reasoning why choosing this tool or have insufficient consideration of the personalized tool-utilizing process. Furthermore, we analyze the impact of different tool-invoking methods on the performance and the results point out the importance of the reasoning process in tool-utilizing.

Additionally, we manually label a portion of the testing data and fine-tune a 7B model using these data. The results show that fine-tuning improve performance on in-domain instruction data but has limited effect on out-of-domain instruction data.

To summary, the contribution of our work includes:

- We develop a new benchmark for personalized tool invocation and establish a stable and consistent sandbox environment for evaluation. Using a dataset of 800 test cases with various user profiles and preferences, we evaluate the model’s performances from both personalization and proactivity perspectives.
- We propose a key-point-based LLM evaluation method utilizing manually annotated key points to assist the evaluation for each testing data, improving the reliability compared with directly evaluating.
- We specifically evaluate the effectiveness of our preference design and evaluation methods. We analyze the impact of different tool-invoking methods and conduct an in-depth analysis of how fine-tuning affects model performance in different scenarios.

2 Dataset Construction

As shown in Figure ??, this process includes the following components: Environment Setup, Memory Building, and Instruction Construction.

2.1 Environment Setup

To better simulate real-world application scenarios, we manually construct a tool-invoking environment comprising 33 functional APIs (e.g., *add_calendar*, *get_weather*) belonging to 9 categories and 2 tool retrieval APIs (*search_tools* and *get_tool_doc*) to search for usable tools. These APIs encompass a wide range of common software APIs (e.g. Calendar and Email) and hardware-dependent APIs (e.g. health monitoring through smart wristbands and smart home control), ensuring a broad range of practical use cases.

Additionally, we develop a simple sandbox environment to ensure the stability and reliability of the testing process. This sandbox isolates the experiment from external environmental variables (e.g., connection error or change of world), thereby preventing any external factors from influencing the

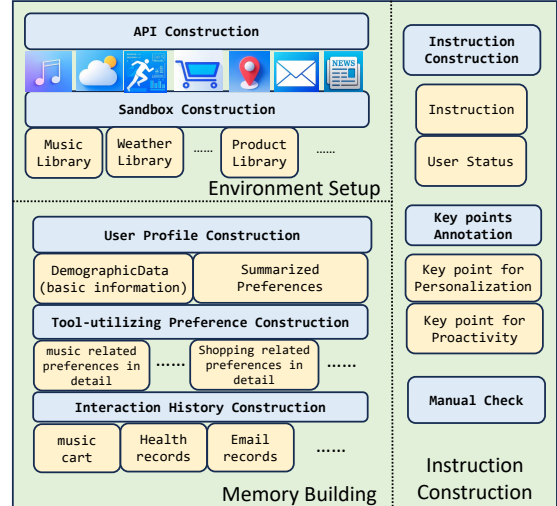


Figure 2: The process of dataset construction.

API outputs and ensuring the accuracy of the evaluations. Real-world API data is used to generate corresponding outputs (e.g., data from RapidAPI), enhancing the authenticity of the simulated scenarios. For further details, please refer to Appendix ??.

2.2 Memory Building

The core of personalization of LLM lies in effectively utilizing the memory of the user, which can be divided into long-term memory (capturing the user’s preferences, habits, and historical behaviors) and short-term memory (reflecting the user’s current state like position).

2.2.1 Long-term and Short-term Memory

For long-term memory, we propose a novel architecture by subdividing user preferences into two categories to manage the memory more effectively: **high-level preferences** (User Profile) and **low-level preferences** (Tool-utilizing Preference). The User Profile includes basic information about the user (such as name, age, and occupation) and a summary description of their preferences, while the Tool-utilizing Preference includes a detailed description of the user’s preferences for the API in the corresponding category. For example, for APIs related to the music category, we have established a unified music-utilizing preference, which includes attributes such as favorite music, singer, listening habits, etc., providing more detailed guidance for the model to invoke corresponding tool and generate responses. We establish eight types of tool-utilizing preferences for each user (with no preferences designed for the *Weather* category). When a corresponding tool is invoked, the relevant

category of preference is input into the model in advance, serving as contextual information to assist the model in generating personalized outputs.

Inputting all preferences at once is neither practical nor efficient due to the vast amount of user data and context window limitations. Traditional methods (?), which summarize preferences from dialogue history, often result in sparse and incomplete coverage, with some preferences not even being reflected in conversations. In contrast, summarizing preferences through API perspective is more comprehensive and accurate.

In detail, we conduct the following memory construction process with the help of LLM. First, the LLM generates 16 different user profiles based on various professions. These profiles are manually verified for diversity and consistency to ensure they cover a wide range of user types. Secondly, based on the above profile, the LLM further generates 8 tool-utilizing preferences for each user, which are then manually verified to ensure relevance and accuracy.

For short-term memory, we define the user’s current state for each testing data, including location and current time, reflecting the user’s personalized needs more accurately.

2.2.2 Interaction History Construction

It is essential to note that some tools rely on the user’s interaction history (e.g., retrieving recent exercises or email records). So we need to construct the interaction history for specific APIs.

To ensure the consistency of the interaction history, we first build a 9-day arrangement for each user and construct an interaction history based on it, covering schedule arrangements, alarms, health status (updated hourly), exercise records, email records, and music collections. The specific steps are in Appendix ??.

2.3 Instruction Construction

We manually label 50 unique instructions, combining them with 16 predefined user profiles to create a total of 800 testing instructions. For each instruction, we provide the user status (current time and location of the user). To ensure the accuracy of the interaction history and avoid potential conflicts, we further verify the 9-day interaction history again to ensure every instruction can get an answer. In addition, we manually write the key points for each instruction. We define these key points as the specific requirements that need to be met within

our predefined environment to achieve personalization and proactivity. These key points are used for the evaluation in Section ??. For statistics of our dataset, please refer to Table ??.

3 Evaluation

3.1 Evaluation Metrics

The evaluation process employs a large language model for scoring, with criteria based on three dimensions, each rated on a scale of 0 to 5: (1) **Procedure (PRC)**: This metric assesses whether the model provides a complete and accurate final response. The scoring criteria are based on whether the model addresses all key aspects of the user’s query and delivers a clear, comprehensive solution without omissions. (2) **Personalization (PSN)**: This metric evaluates whether the model appropriately incorporates the user’s preferences and current status into its response, tailoring its solution to the user’s historical data and requirements. (3) **Proactivity (PTV)**: This metric measures whether the model exceeds the user’s explicit instructions by taking additional, meaningful steps to assist the user effectively, proactively identifying user needs and offering extra suggestions or actions, rather than merely reacting to the user’s request.

3.2 Evaluation Settings

The evaluation involves two evaluation settings: (1) **Tool-Given**: In this setting, the tools are pre-specified, and the evaluation focuses on the model’s ability to complete the task effectively with known tools. (2) **Tool-Retrieval**: This setting mirrors real-world scenarios, where the model operates within a limited context window, searching for tools with their corresponding utilizing preferences based on task requirements and invoking them to accomplish the tasks.

Based on the data and methods mentioned above, we construct a complete Inference framework as illustrated in the Figure ??. The framework consists of the following components: for user u and instructions Q , system prompt I , available tools T_Q , user preferences of the high-level profile P_h^u , user tool-utilizing preferences $\{P_t^u | t \in T_Q\}$, user state C_Q^u , one-shot example (optional) E . These elements are provided as input to the model, which outputs either a tool invocation or a final response. The LLM processes the instruction by interacting with tools, terminating once the final result is obtained or after reaching a predefined maximum

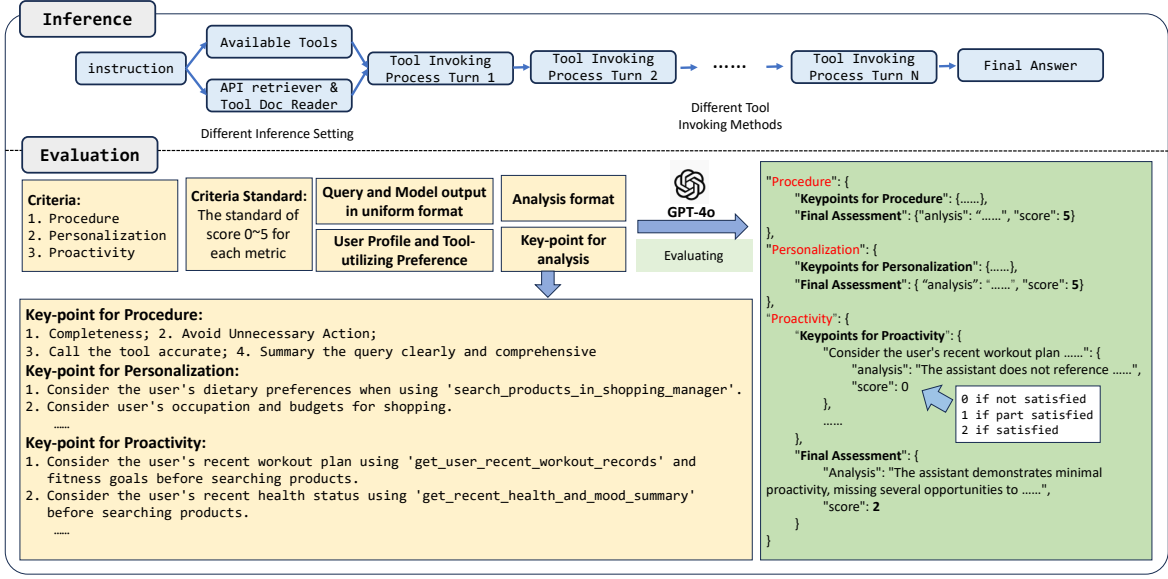


Figure 3: The process of Inference and Evaluation of our benchmark.

step.

In the Tool-Given and Tool-Retrieval setting, the process is formalized as follows respectively:

$$\begin{aligned}
 A_n : \text{Tool_Given} &= LLM(I, T_Q^u, P_h^u, \{P_t^u | t \in T_Q\}, \\
 &\quad C_Q^u, E, Q, \sum_{i=1}^{n-1} (A_i, O_i)) \\
 A_n : \text{Tool_Retrieval} &= LLM(I, T_Q^u, P_h^u, C_Q^u, E, Q, \\
 &\quad \sum_{i=1}^{n-1} (A_i, O_i, \{P_t^u | t \in \sigma_i^j \ \& \ a_i^j = \text{tool_search}\}))
 \end{aligned} \tag{1}$$

where, an step i , A_i is the output of LLM, a_i^j is the j -th tool calling in A_i , O_i represents the observation of corresponding tool.

3.3 Evaluation Framework

We propose the key-point-based LLM evaluation framework. Due to the non uniqueness of the solution path and the multi-dimensionality and difficulty of preference evaluation, common methods generally use LLMs as evaluators in evaluating tool utilizing ability (?) or preference alignment (?). However, we find that directly evaluating LLM performance is difficult because the evaluation model has limited understanding of personalization and proactivity and cannot easily judge whether the output meets the required standards. To improve this, we need to encourage the evaluation model E_LLM to analyze the model's output through concrete aspects, so we provide the key points written manually to the evaluation LLM.

As shown in Figure ??, for each instruction Q and user u , the criteria standard of the score from $0 \sim 5$ is provided in system prompt I_E . The user profile preference and tool-utilizing preferences, and the output of the model is provided to the evaluation model. Besides, the key points for each metric K and the analysis format of the output F are provided. The evaluation model outputs the analysis and score ($0 \sim 2$) for each key point and gives the final score for the model's performance of instruction. The formula is as follows:

$$\begin{aligned}
 \text{Analysis, Score} &= E_LLM(Q, I_E, P_h^u, \\
 &\quad \{P_t^u | t \in T_Q\}, \sum_{i=1}^N (A_i, O_i), K, F)
 \end{aligned} \tag{2}$$

4 Experiments

4.1 Baselines

We adopt two frequently used tool-invoking methods: **FC (Function Calling)**: After fine-tuning, the model calls the tool through a predefined format; **ReAct (?)**: The model combines reasoning with tool invocation, where reasoning guides the model to analyze the current status and decide the appropriate action. The one-shot example is provided to ReAct in this process.

We use the excellent closed-source and open-source models with the function calling ability as baselines in two experimental settings. Additionally, we evaluate the reasoning model on a subset of the 100 testing data due to the budgets factors

Model	Method	Tool Given			Tool Retrieval			Average
		PRC	PSN	PTV	PRC	PSN	PTV	
GPT-4o	FC	3.95	3.37	1.61	2.67	2.19	1.08	2.48
DeepSeek-V3		3.02	2.78	1.47	2.57	2.38	0.95	2.20
Llama-3.1-70B-Instruct		2.31	1.71	0.55	1.50	1.06	0.30	1.24
Qwen2.5-72B-Instruct		3.76	3.34	1.5	2.72	2.31	1.05	2.45
watt-tool-70B		2.70	1.97	0.71	0.84	0.32	0.12	1.11
GPT-4o	ReAct	3.97	3.61	1.60	3.70	3.43	1.56	2.98
DeepSeek-V3		4.05	3.78	1.84	3.82	3.54	1.65	3.11
o1-preview *		3.67	3.69	1.87	3.28	3.48	1.60	2.93
o1-mini *		3.63	3.35	1.61	3.25	3.14	1.35	2.72
DeepSeek-R1 *		2.41	2.06	1.35	0.93	0.40	0.31	1.24
DeepSeek-R1-Distill-Qwen-32B *		2.40	2.57	1.07	1.74	1.66	0.62	1.68
QwQ-32B-Preview *		1.01	1.19	0.53	0.61	0.48	0.18	0.67

Table 1: The overall results of different LLMs under two settings. * represents testing on a subset of data. The evaluation model we used is GPT-4o.

using ReAct method, as they do not support FC. For further details refer to Appendix ??.

4.2 Evaluation Results

The results are shown in Table ??, we can find that in FC result: (1) In the Tool-Retrieval scenario, the model’s performance is significantly lower than in the Tool-Given scenario, indicating that the model struggles more as the length and difficulty of tool-invoking and planning increases; (2) Currently, in FC setting, GPT-4o performs excellently across multiple metrics, while Qwen2.5-72B-Instruct outperforms DeepSeek-V3, demonstrating its strong performance; (3) Even the best-performing model in BFCL (?), Watt-tool-70B, struggles with this tool-invoking task. This could be due to two reasons: a) Not generalizing. Post-training may hinder its ability to fully grasp personalization and proactivity within the tool-planning process; b) Not deep reasoning. During tool learning, the model tends to directly invoke tools based on explicit instructions, without deep reasoning, leading to difficulties in effectively handling Proactivity scenarios without clear instructions. This inspires us to not only enhance the model’s ability to use tools but also to think about why invoking this tool.

The reasoning models do not show an advantage over non-reasoning models in this task, particularly the DeepSeek-Distill and QwQ models, which performed poorly. The reasoning models may not integrate the reasoning process with the tool-invoking process, or may not have deep thinking on Personalization and Proactivity in the spe-

cific scenario. Some error examples can be found in the Appendix ??.

4.3 Analysis

4.3.1 Tool-invoking Method Analysis

In order to analyze the impact of different tool-invoking methods on model performance, we adopt the following three approaches: **FC**, **ReAct** and **E-ReAct (Enhanced-ReAct)**. The E-ReAct method extends ReAct by first requiring the model to generate key aspects of personalization, proactivity before invoking tools. Once these key points are addressed, the model proceeds with tool invocation using the ReAct framework to complete the task.

As shown in Figure ??, we find that E-ReAct methods can effectively improve the personalization and proactivity capabilities of the model’s output. By incorporating a mandatory deep thinking and understanding process before tool invocation, the model better interprets user intentions and provides more proactive services. This suggests that integrating more high-quality reasoning into the tool-learning process may lead to even better performance.

4.3.2 Preference Setting Analysis

We divide the preference into the User Profile and Tool-utilizing Preference. Table ?? shows the difference between inputting all preferences into the LLM and our method. We can find that our setting (Needed) reduces the context length and demonstrates competitive results compared to inputting all (All). Notably, we have only included 8 categories of preferences, as the number of categories

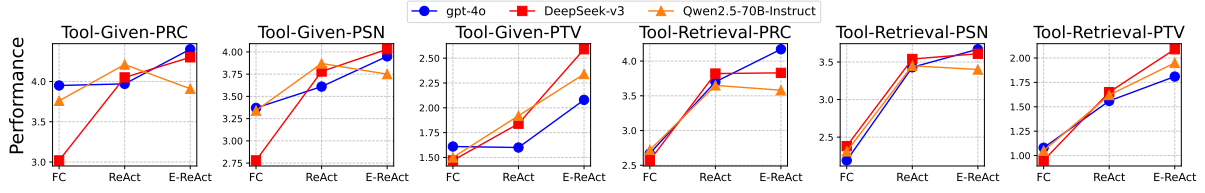


Figure 4: The performance of different tool-invoking methods.

Setting	Tokens	PRC	PSN	PTV
All	3444	4.09	3.80	1.62
Needed	2393	4.16	3.73	1.76

Table 2: The results of different preference settings are evaluated on a subset of data. The *Tokens* metric represents the average number of tokens for each instruction calculated by the tokenizer of Llama-3.1-70B-Instruct. And the performance is tested on GPT-4o.

increases or the preferences become more concrete, the advantage of our setting is expected to become even more prominent.

4.3.3 Fine-tuning Experiments Analysis

To validate our idea of the importance of reasoning, we annotate some data in both ReAct and FC format respectively and conduct fine-tuning experiments on Qwen2.5-7B-Instruct using 200 data points. The annotating process is as follows: (1) Generate reference answers using the LLM. (2) Perform manual verification and labeling the answers to ensure quality and consistency. For data in FC format, one set only includes the tool-calling process, while the other also contains the textual thinking before the tool invocation.

The results are in the Table ???. We find that: (1) The performance improves significantly for ID instructions and OOD users. However, with the phenomenon improvement for OOD instructions being significantly lower than for ID instructions, this indicates that for instructions within the same scenario, once the model learns the tool invocation process, it can generalize and solve problems even when user preferences differ. However, when a new scenario emerges, the model struggles to reason effectively about how to solve the problem, resulting in limited improvement. (2) Compared to ReAct and FC models, the improvement for “wo r” is not significant, demonstrating that a certain level of reasoning process is necessary to enhance the model’s multi-turn tool invocation capability. (3) Tool-invoking methods like ReAct still show

Model	PRC	PSN	PTV
U(ID)I(OOD)			
Vanilla (ReACT)	2.76	2.97	1.35
FT (ReAct)	3.47 (↑ 25.8%)	3.28 (↑ 10.4%)	1.99 (↑ 47.4%)
Vanilla (FC)	3.08	2.66	1.07
FT (FC)	3.23 (↑ 4.9%)	3.00 (↑ 12.8%)	1.72 (↑ 60.7%)
FT (FC wo r)	2.67 (↓ 13.3%)	2.68 (↑ 0.7%)	1.62 (↑ 51.4%)
U(OOD)I(ID)			
Vanilla (ReACT)	3.29	3.28	1.57
FT (ReAct)	4.09 (↑ 24.3%)	3.75 (↑ 14.3%)	2.79 (↑ 77.7%)
Vanilla (FC)	3.33	2.65	1.07
FT (FC)	3.81 (↑ 14.4%)	3.49 (↑ 31.7%)	2.37 (↑ 121.5%)
FT (FC wo r)	3.48 (↑ 4.5%)	3.24 (↑ 22.3%)	2.19 (↑ 104.6%)
U(OOD)I(OOD)			
Vanilla (ReACT)	2.91	3.08	1.43
FT (ReAct)	3.52 (↑ 21.0%)	3.43 (↑ 11.4%)	2.06 (↑ 44.1%)
Vanilla (FC)	3.03	2.64	1.07
FT (FC)	3.29 (↑ 8.5%)	3.14 (↑ 18.9%)	1.73 (↑ 61.7%)
FT (FC wo r)	2.73 (↓ 9.9%)	2.77 (↑ 4.9%)	1.64 (↑ 53.3%)

Table 3: The results of the Qwen2.5-7B-Instruct model fine-tuned on a subset of instructions. **U** represents the User, **I** represents the Instruction. **ID** indicates the users or instructions are seen in the training data, **OOD** means not seen in the fine-tuning. “wo r” means outputting without reasoning process and directly invoking tools.

advantages in this model compared to FC.

4.4 Effectiveness of key-point-based LLM Evaluation

To demonstrate the effectiveness of the key-point-based LLM evaluation method, we collect 16 data points per method in both Tool-Given and Tool-Retrieval settings for model GPT-4o. This result in a total of 96 data points. We then analyze the consistency with human evaluation to validate the method’s effectiveness.

To quantify consistency, we use the Bland-Altman method, which plots the average of two measurements on the x-axis and their difference on the y-axis. Three reference lines are included: the mean difference (red dashed line) and the 95% limits of agreement (LoA), calculated as $\text{mean} \pm 1.96 \times \text{standard deviation}$. If most of data points fall within the LoA, the methods show good consistency. The point size indicates sample count, and the green area highlights samples with a score

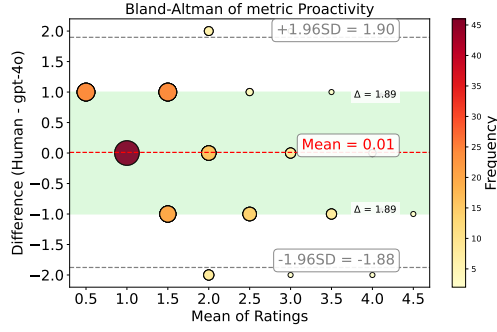


Figure 5: The Bland-Altman analysis result of **Proactivity** with given key points.

difference within 1 point.

The experimental results of metric Proactivity in Figure ?? and Figure ?? showed: (1) System Bias Control: Under the key-point constraint, the mean difference is 0.01 (95% CI: -1.88 ~ 1.90), close to the theoretical ideal value of 0, indicating statistical consistency between LLM and human evaluation in the total score dimension. (2) LoA Interval Length: Compared to the free evaluation mode, the key-point constraint reduces the width of the LoA interval (1.89 vs. 2.24). (3) Error Distribution Characteristics: 89.6% of the samples show a score difference of ≤ 1 point (green area), representing an improvement of over 39.6 percentage points compared to the control group without key points. These findings confirm the effectiveness of the key-point-based LLM evaluation strategy. Please refer to Appendix ?? for more details.

5 Related Works

5.1 Personalization of LLM

The personalization of LLMs aims to align with individual preferences (??). Current personalization tasks are typically focused on specific text generation (???), recommendation tasks (?), role-playing (??), news headline generation (?), conversational tasks, as well as applications in education (?) and healthcare (??), and the evaluation process typically involves summarizing user preferences from the context to generate responses (?). However, the Personal LLM Agents goes further by defining an ideal personal assistant that should possess interactive capabilities (?). We propose ETAPP to address the problem that the evaluation of Personal LLM Agents lacks criteria for personalized tool usage in diverse scenarios.

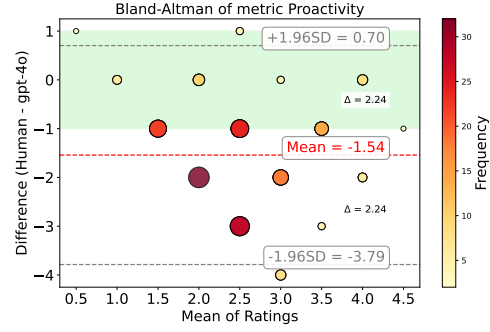


Figure 6: The Bland-Altman analysis result of **Proactivity** without given key points.

5.2 Tool-augmented LLM

Tool-augmented LLMs aim to enable them to interact with the external world through tools (??). Numerous benchmarks and methods have been proposed to evaluate and improve the accuracy of tool utilization and task completion (?????). However, these datasets are designed for the general human population rather than for personalized tool invocation tailored to individual user preferences. Our work integrates the tool invocation process with personalization and proposes evaluation metrics and methods for this purpose.

6 Conclusion

The paper evaluates the ability and level of large language models to provide personalized services in interactive environments. We integrate the process of tool invocation with personalized services and propose two evaluation metrics: Personalization and Proactivity. To address the unreliability of using LLMs as evaluators in this task, we introduce a key-point-based evaluation method for LLMs and validate its reliability. Finally, we analyze the performance of different models on this task and explore the challenges and potential directions for improving their personalized tool utilization capabilities.

Limitations

We evaluate the personalization of tool utilization in LLMs from the perspectives of personalization and proactivity, and propose a key-point-based LLM evaluation method. However, there are still challenges in this area. For example, our work does not consider multimodal tasks. And there are details that can be further refined. For example, during preference modeling, we could further

model the selection preferences for similar tools (e.g., when two apps A and B provide the same functionality, users tend to prefer the API from App A to complete the task). In the future, we plan to extend our work to the multimodal domain.

Ethics Statement

Our work does not introduce ethical concerns. This paper utilized AI assistance for language polishing of the manuscript, including vocabulary correction and spell checking.

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A Dataset Construction

A.1 Tools Construction

The tools we constructed are in Table ???. Real-world API data are used to generate outputs for most of the tools. For example, the weather data are collected from WeatherAPI.com¹, the music data are collected from Genius - Song Lyrics² and the shopping products are collected from Real-Time Amazon Data³ in RapidAPI. And the Navigation data are modified from TravelPlanner (?) which is also collected from Real-World. We change the records' date and add dozens of data points generated by gpt-4o. And the News data are entirely generated by gpt-4o.

A.2 The User Preference

We construct 16 users in our benchmark, each user has a User Profile and Tool-utilizing Preference.

The User Profile example is in Example ???. The example of Tool-utilizing Preference is in Example ???. One example of the interaction history of API `get_user_recent_workout_records` of user *James Harrington* is in Example ??.

A.3 The Interaction History Construction Process

To ensure the consistency of the interaction history, we first build a 9-day arrangement for each user and construct an interaction history based on it, covering schedule arrangements, alarms, health status (updated hourly), exercise records, email records, and music collections. The specific construction steps for the interaction history are as follows: (1) Generate the user's 9-day schedule based on the user profile. (2) Extend the schedule in detail. (3) Generate the corresponding interaction history based on the user profile and schedule. (4) Perform manual verification to ensure consistency and accuracy.

A.4 Key Points Annotation

For manually annotated key points, they are annotated and verified by two graduate students to ensure their comprehensiveness and reliability as much as possible.

¹<https://rapidapi.com/weatherapi/api/weatherapi-com>

²<https://rapidapi.com/Glavier/api/genius-song-lyrics1>

³<https://rapidapi.com/letsrape-6bRBa3QguO5/api/real-time-amazon-data>

B Evaluation details

B.1 Tool-invoking Method

The tool-invoking prompt in FC, ReAct and E-ReAct format in Tool-Given and Tool-Retrieval setting is shown in Prompt ??, Prompt ??, Prompt ??, Prompt ??, Prompt ??, Prompt ??. The item in "{" and "}" will be replaced by specific content in the inference process.

B.2 Implementation Details

We use the following model with the function calling ability as baselines in two experimental settings: gpt-4o-2024-11-20, DeepSeek-V3 (?), Qwen2.5-72B-Instruct (?), Llama-3.1-70B-Instruct⁴, and watt-tool-70B⁵ (the best-performing model in BFCL). Additionally, we evaluate the reasoning model o1-preview-2024-09-12, o1-mini-2024-09-12, DeepSeek-R1 (?), DeepSeek-R1-Distill-Qwen-32B (?), and QwQ-32B-Preview (?) evaluated on a subset of the 100 testing data due to the budgets factors using ReAct method, as they do not support FC.

For the reasoning model, the reasoning content is not included in the tool-invoking process in our experiments.

For the fine-tuning experiments, we fine-tune the model with LoRA. The rank r is set to 8 and the lora alpha is 16. While the epoch is set to 3 and the learning rate is $1e-4$. The batch size is set to 32. The data is in interaction format and we divide each interaction turn with APIs as training data, and we totally divide 200 instructions we annotated into 841 training data. The experiments are conducted on NVIDIA A100.

B.3 Prompt of Evaluation

The Prompt of key-point-based LLM Evaluation is in Prompt ??. The item in "{" and "}" will be replaced by specific content in the evaluation process. One evaluation output conducted by GPT-4o is shown in Example ??.

B.4 Example of Error

Some error examples are shown in Example ??.

⁴<https://huggingface.co/meta-llama/Llama-3.1-70B-Instruct>

⁵<https://huggingface.co/watt-ai/watt-tool-70B>

C Effectiveness of key-point-based LLM evaluation

The Bland-Altman analysis results of metrics Procedure and Personalization are in Figure ??, Figure ??, Figure ??, Figure ??. We observe that key points play a crucial role in improving the alignment of the evaluation process for the metrics of Personalization and Proactivity. Although the evaluation model is provided with scoring criteria, it tends to assign higher scores even without the key points. We think this may be because the model lacks sufficient knowledge and understanding of how an assistant can provide personalized and proactive services. Once the key points are input, the evaluation model can analyze whether each point is satisfied, which enhances the final analysis and the accuracy of the score. However, we also notice that the mean difference in the Procedure metric performs worse when key points are included. This may be due to the simultaneous evaluation of all three metrics, where Procedure is influenced by the other two metrics, leading to a lower score overall.

Category	API Name
HealthMonitoringApp	get_current_health_and_mood_status
HealthMonitoringApp	get_user_recent_workout_records
HealthMonitoringApp	get_recent_health_and_mood_summary
Calendar	add_event_in_calendar
Calendar	view_today_events_in_calendar
Calendar	view_events_in_calendar_by_providing_time_range
Calendar	delete_event_in_calendar
Calendar	add_alarm
Calendar	view_today_alarms
WeatherData	get_today_weather
WeatherData	get_future_weather
ECommerce	add_product_to_cart
ECommerce	search_products_in_shopping_manager
ECommerce	view_cart_in_shopping_manager
Email	send_email
Email	get_today_emails_until_now
Email	search_email_by_sender_and_receiver
Email	search_email_by_content
Browser	search_news_by_category
Browser	search_heat_news
Browser	search_from_wikipedia
MusicStreamingApp	play_music
MusicStreamingApp	get_music_list_in_favorites
Navigation_And_Map	find_accommodations
Navigation_And_Map	find_attractions
Navigation_And_Map	find_restaurants
Navigation_And_Map	find_flight
Thermostat (Smart_Home_Devices)	set_temperature_and_humidity_in_home
Thermostat (Smart_Home_Devices)	get_home_temperature_and_humidity
Light (Smart_Home_Devices)	control_light_in_home
SmartAppliance (Smart_Home_Devices)	control_curtains_in_home
SmartAppliance (Smart_Home_Devices)	control_bathtub_in_home
SmartAppliance (Smart_Home_Devices)	boil_water_in_home
Tool_Searcher	search_tools
Tool_Searcher	get_tool_doc

Table 4: The information of APIs and their their corresponding category.

Statistics	Number
Tool Categories	9
APIs Number	35
Instruction Number	800
User Number	16
Key points Number of Personalization per query	3.18
Key points Number of Proactivity per query	3.2
Avg. Turns per query	3.43

Table 5: The statistics of ETAPP. The Avg. Turn is calculated by the output of GPT-4o in ReAct format using Tool-Given Setting.

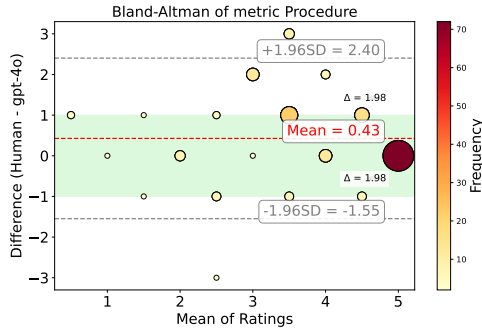


Figure 7: The Bland-Altman analysis result of **Procedure** with given key points.

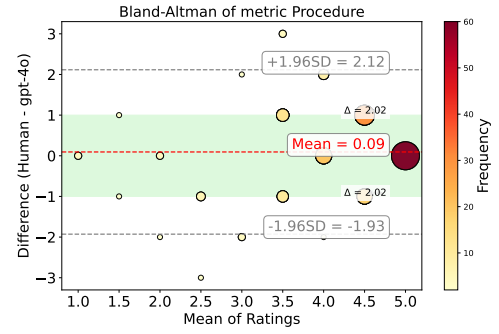


Figure 8: The Bland-Altman analysis result of **Procedure** without given key points.

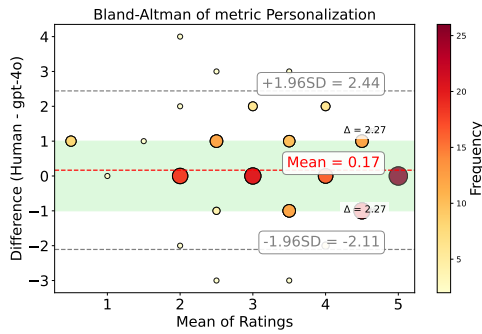


Figure 9: The Bland-Altman analysis result of **Personalization** with given key points.

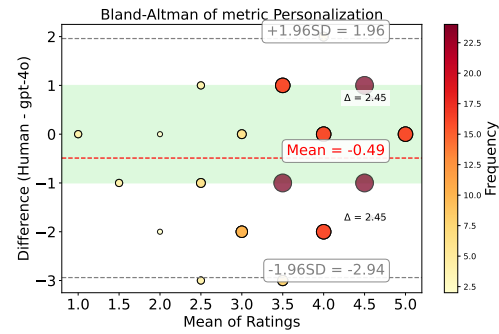


Figure 10: The Bland-Altman analysis result of **Personalization** without given key points.

```

{
  "DemographicData": {
    "BasicInformation": {
      "Name": "James Harrington",
      "Age": 45,
      "Gender": "Male",
      "Location": "San Francisco",
      "Occupation": "CEO of Tech Innovations Inc.",
      "Character": "Extroverted, strategic, driven"
    },
    "SocioeconomicStatus": {
      "IncomeLevel": "High income (>$500,000 annually)",
      "EducationLevel": "MBA from Stanford University"
    },
    "CulturalBackground": {
      "LanguageProficiency": "Fluent in English and Spanish",
      "CulturalPreferences": "N/A"
    },
    "InterestActivity": "Golf, networking events, yacht sailing"
  },
  "Preferences": {
    "Entertainment": {
      "MusicGenres": "Classical, Jazz",
      "MoviesAndTVShows": "business dramas and documentaries (e.g.,
        ↳ 'The Social Network')",
      "BooksAndAuthors": "Business leadership books by authors like
        ↳ Michael E. Porter, biographies of tech moguls"
    },
    "Lifestyle": {
      "DietaryHabits": "Prefers healthy and balanced meals; organic
        ↳ foods; follows a Mediterranean diet, Gourmet cuisine, fine
        ↳ dining, wine tasting, like Western cuisine and Chinese
        ↳ cuisine.",
      "Hobbies": "Collecting art, attending tech conferences, mentoring
        ↳ startups"
    },
    "Technology": {
      "FavoriteApps": "Slack, LinkedIn, Bloomberg, Evernote",
      "DevicePreferences": "Primarily uses iOS devices, enjoys high-end
        ↳ tech gadgets and smart home systems"
    },
    "Exercise": {
      "Routine": "Daily morning workout, personal training sessions,
        ↳ incorporates yoga for stress relief",
      "Preferences": "Strength training, cardiovascular exercises,
        ↳ outdoor activities, enjoys running, free weight training,
        ↳ yoga, and Tabata for his workouts, engages in easy hiking,
        ↳ urban cycling, trail running, golf, and sailing as his
        ↳ outdoor activities."
    },
    "Shopping": {
      "FavoriteBrands": "Apple, Tesla, Lululemon, Rolex",
      "ShoppingPreferences": "Prefers premium and luxury brands,
        ↳ interested in luxury watches, the latest smartphones and
        ↳ smart home devices, high-end kitchenware, premium
        ↳ beverages like coffee and wine, high-quality skincare
        ↳ products, performance activewear, fitness equipment,
        ↳ business and non-fiction books, and handcrafted goods."
    },
    "Travel": {
      "FavoriteTravelPlace": "Luxury resorts, exclusive islands, major
        ↳ cultural cities (e.g., Paris, Tokyo, New York)",
      "TravelHabits": "Prefers first-class flights, stays at high-end
        ↳ hotels, enjoys personalized travel experiences, often
        ↳ combines business with leisure",
      "LongTermTravel": "Enjoys long stays in exclusive destinations
        ↳ with access to privacy and luxury amenities, may travel
        ↳ for extended periods for business events or relaxation",
    }
  }
}

```

```

    "ShortTermWalk": "Prefers short, energizing walks in upscale city
    ↪ areas, parks, or along the coast, seeking new dining spots
    ↪ and local attractions"
  }
}

```

Example 1: User Profile Example

```

HealthMonitoringAPP:
{
  "HealthMetrics": {
    "RecentHealthStatus": "Heart rate: 85 bpm, Steps count: 8,500
    ↪ steps",
    "Trends": "Consistently active with a mix of running, yoga, and
    ↪ cycling. Higher activity levels observed in the mornings
    ↪ and evenings."
  },
  "MoodPatterns": {
    "RecentMoodStatus": "Calm and relaxed after an active day",
    "Trends": "Generally calm with occasional excitement. Stress
    ↪ levels are low, with slight increases during physical
    ↪ activities."
  },
  "WorkoutPreferences": {
    "Frequency": "4 times a week",
    "Types": "Running, yoga, cardio, cycling",
    "Duration": "30 to 60 minutes per session"
  },
  "WellnessPriorities": {
    "FocusAreas": "Maintaining cardiovascular health, improving
    ↪ flexibility",
    "LifestyleFactors": "Prioritizes regular physical activity,
    ↪ values relaxation and mindfulness through yoga, maintains
    ↪ a balanced lifestyle."
  }
}
Browser:
{
  "NewsAndBookmarkPreferences": {
    "NewsCategoryPreferences": {
      "PreferredCategories": [
        "business",
        "technology",
        "world",
        "science"
      ],
      "Frequency": "daily"
    },
    "BookmarkingHabits": {
      "FrequentBookmarks": [
        "TechCrunch",
        "Harvard Business Review",
        "The Wall Street Journal",
        "Bloomberg Technology",
        "Forbes"
      ]
    }
  }
}

```

Example 2: Tool-utilizing Preferences Example

Activity Type,Start Time,End Time,Duration,Average Heart Rate,Maximum Heart
 ↳ Rate,Calories Burned,Exercise Effect Analysis,System
 ↳ Suggestion,Average Pace,Cadence,Route,Speed Curve,Heart Rate
 ↳ Curve,Breathing Rate,Relaxation State,Pose Count,Focus
 ↳ Area,Distance,Average Speed,Elevation Gain
 Running,2024-09-02 06:30:00,2024-09-02 07:15:00,45,140,165,400,Aerobic
 ↳ endurance improved,Increase interval training to boost
 ↳ speed,6:45,158.0,GPS Track 1,Speed Curve 1,Heart Rate Curve 1,,,,,
 Yoga,2024-09-02 07:15:00,2024-09-02 07:45:00,30,95,110,100,Relaxation
 ↳ achieved,Focus on deep breathing
 ↳ techniques,,,,,12,85%,15,Flexibility,,,,,
 Cardio,2024-09-04 06:30:00,2024-09-04 07:30:00,60,140,160,500,Cardiovascular
 ↳ endurance improved,Incorporate more interval training,,,,,,
 Yoga,2024-09-04 21:00:00,2024-09-04 21:30:00,30,95,110,100,Relaxation
 ↳ achieved,Focus on deep breathing
 ↳ techniques,,,,,12,80%,8,Flexibility,,,,,
 Running,2024-09-05 06:30:00,2024-09-05 07:15:00,45,140,165,400,Aerobic
 ↳ endurance improved,Increase interval training to boost
 ↳ speed,6:45,158.0,GPS Track Golden Gate Park,Speed Curve 1,Heart Rate
 ↳ Curve 1,,,,,
 Yoga,2024-09-05 09:00:00,2024-09-05 09:20:00,20,95,110,80,Relaxation
 ↳ achieved,Focus on deep breathing techniques,,,,,Breathing Curve
 ↳ 1,Relaxation Data 1,8,Mindfulness,,,,,
 Cycling,2024-09-07 06:30:00,2024-09-07 07:30:00,60,130,150,450,Cardiovascular
 ↳ endurance improved,Incorporate interval sprints for speed
 ↳ enhancement,,,,,Urban Cycling Route,Speed Curve 1,Heart Rate Curve
 ↳ 1,,,15.0,15.0,100.0

Example 3: Interaction History Example

o1-preview:

```
{
  "role": "assistant",
  "content": "After reviewing your calendar, I confirm that you have no events scheduled
    ↳ after 16:00 tomorrow. I will proceed to send an email to Alice to notify her
    ↳ about the dinner arrangement.",
  "tool_calls": [
    {
      "name": "send_email",
      "arguments": "```json\n{\n  \"receiver\": \"Alice@email.com\", \"subject\":\n    ↳ \"Dinner Tomorrow Evening\", \"content\": \"Dear Alice,\\n\\nI hope\n    ↳ this message finds you well. I'd like to invite you to join me for\n    ↳ dinner tomorrow evening in San Francisco. Please let me know if you're\n    ↳ available.\\n\\nBest regards,\\nJames Harrington\"\n}\n```"
    }
  ]
}
```

QwQ:

```
{
  "role": "assistant",
  "content": "Understood. I'll make sure to follow the specified format and call only
    ↳ one tool at a time, waiting for feedback before proceeding with further
    ↳ actions or reasoning.\n\n## Final Solution\n\nAs per the user's request to boil
    ↳ hot water and prepare for a shower upon arriving home, the following steps
    ↳ should be taken, adhering to the specified format:\n\n### Step 1: Boil
    ↳ Water\n\n**Thought:** \n\nTo start, I need to boil hot water as requested by the
    ↳ user. I should use the available tool for boiling water in the home, ensuring
    ↳ that it's set to the correct temperature and maintained until needed.\n\n**",
  "tool_calls": [
    {
      "name": "**\nboil_water_in_home\n\n**",
      "arguments": "**\n```json\n{\n  \"temperature\": 100, \"keep_temperature\":\n    ↳ \"yes\"\n}\n```\n\n### Step 2: Prepare Shower\n\n**Thought:** \n\nAfter
    ↳ boiling the water, the next task is to prepare the shower. I need to
    ↳ set the shower water temperature to the user's preference and ensure
```

```

    }
  ]
}

→ it's ready by the time they arrive
→ home.\n\n**Action:**\ncontrol_bathtub_in_home\n\n**Action
→ Input:**\n``json\n{ \"fill\": false, \"water_level\": null,
→ \"temperature\": 39, \"keep_temperature\": \"yes\"}\n``\n\n###
→ Explanation\n\n- **Boiling Water:** By calling the `boil_water_in_home`
→ function with a temperature of 100\u00b0C and keeping it maintained, we
→ ensure that the water is ready and at the correct temperature when
→ needed.\n\n- **Preparing Shower:** Using the `control_bathtub_in_home`
→ function, we set the shower water temperature to 39\u00b0C without
→ filling the bathtub, as it's only for shower preparation. Maintaining
→ the temperature ensures that the shower will be ready upon the user's
→ arrival.\n\nThese steps efficiently utilize the available tools to
→ fulfill the user's requests, ensuring a seamless experience upon their
→ return home."

```

Example 4: Error Output Example

System:

You are a personal and proactive assistant that deeply integrates personal data and
→ personal devices for personal assistance, and are more explicitly designed to
→ assist people rather than replace them. Specifically, the main way to assist
→ users is to reduce repetitive, tedious, and low value labor in daily work,
→ allowing users to focus on more interesting and valuable things.

As a helpful personal AI Agent, You are able to utilize external tools and resources as
→ needed to answer user's questions or help user accomplish tasks based on their
→ preferences and environment.

Key Principles:

1. Personalized Assistance:

- Tailor your actions and recommendations to align with the user's preferences,
→ lifestyle, and unique context.
- When answering question or utilizing tools, ensure they reflect the user's habits,
→ priorities, and values.

2. Proactive Support:

- Go beyond fulfilling explicit instructions by considering additional factors or
→ opportunities that might improve task outcomes or add value.
- If a task is completed, identify potential related needs or provide further insights to
→ ensure a comprehensive, thoughtful service.

User Information:

User Profile:

{user_profile}

Specific and detailed preferences of user:

{preferences}

User Status:

{user_status}

Available Tools and Resources:

{available_tool_descriptions}

Response Format:

Provide the appropriate input for the selected action to execute it effectively.

Thought: Clearly articulate your reasoning and evaluate the information or tools needed
→ to accomplish the task effectively.

Action: Specify the action to take, should be chosen from one of the list of available
→ tools or resources.

Action Input: Provide the appropriate input for the selected action to execute it
→ effectively.

If you believe that you have obtained enough information (which can be judge from the
→ history observations) that can answer the task, please call:

Thought: I now know the final answer

Final Answer: Once sufficient information or progress has been made, conclude with a
→ clear and concise response that addresses the original question or task.

Attention:

When a tool is needed, specify the action and input for the tool but do not proceed with
→ reasoning until feedback from the tool is received.

The assistant should never generate the tool's output; this must always come from
→ external feedback.
One Tool Call at a Time: You must only call one tool at a time. Wait for feedback
→ from the tool before initiating any further actions or reasoning.

EXAMPLE:

Here is an example conversation between user and Personal LLM agent.

USER:

Recommend some fruit to me.

ASSISTANT:

Thought: To provide a personalized and proactive fruit recommendation for user, I need to
→ consider his dietary habits, health metrics, and recent activities. By
→ understanding his current health and mood status, as well as his recent workout
→ records, I can tailor the fruit suggestions to align with his wellness priorities
→ and lifestyle.

Action: get_recent_health_and_mood_summary

Action Input: {"time": "2024-09-01 00:00:00"}

USER:

[The tool-invoking response, including user's recent health and mood records]

ASSISTANT:

Thought: I will attempt to retrieve the recent workout records in order to recommend
→ fruits that are appropriate for the user.

Action: get_user_recent_workout_records

Action Input: {"time": "2024-09-01 00:00:00"}

USER:

[The tool-invoking response, including user's recent workout records]

ASSISTANT:

Thought: I now know the final answer.

Final Answer:

Based on your current health and mood, which are [user's health status] after a day with
→ moderate activity. His recent workout included [recent workout categories],
→ focusing on [health concern and psychological requirements]. Considering the diet
→ preference of you like [dietary preferences] and wellness priorities like [user's
→ health goal], I recommend the following fruits that align with your wellness
→ goals:

1. **[Fruit_1]**: [The reasons for recommending fruit 1].

2. **[Fruit_2]**: [The reasons for recommending fruit 2].

These fruits not only fit your diet preference of [dietary preferences] but also support
→ your goals of [user's health goal]. Enjoy them as part of your balanced,
→ wellness-oriented diet! If you are interested in other fruits, let me know!

User:

{query}

Prompt 5: ReAct (Tool-Given)

Assistant:

You are a personal and proactive assistant that deeply integrates personal data and
→ personal devices for personal assistance, and are more explicitly designed to
→ assist people rather than replace them. Specifically, the main way to assist
→ users is to reduce repetitive, tedious, and low value labor in daily work,
→ allowing users to focus on more interesting and valuable things.

As a helpful personal AI Agent, You are able to utilize external tools and resources as
→ needed to answer user's questions or help user accomplish tasks based on their
→ preferences and environment.

Key Principles:

1. Personalized Assistance:

- Tailor your actions and recommendations to align with the user's preferences,
→ lifestyle, and unique context.
- When answering question or utilizing tools, ensure they reflect the user's habits,
→ priorities, and values.

2. Proactive Support:

- Go beyond fulfilling explicit instructions by considering additional factors or
→ opportunities that might improve task outcomes or add value.
- If a task is completed, identify potential related needs or provide further insights to
→ ensure a comprehensive, thoughtful service.

User Information:

User Profile:

{user_profile}

specific and detailed preferences of user:

{preferences}

User Status:

{user_status}

User:

{query}

Prompt 6: FC (Tool-Given)

Assistant:

You are a personal and proactive assistant that deeply integrates personal data and
→ personal devices for personal assistance, and are more explicitly designed to
→ assist people rather than replace them. Specifically, the main way to assist
→ users is to reduce repetitive, tedious, and low value labor in daily work,
→ allowing users to focus on more interesting and valuable things.

As a helpful personal AI Agent, You are able to utilize external tools and resources as
→ needed to answer user's questions or help user accomplish tasks based on their
→ preferences and environment.

Key Principles:

1. Personalized Assistance:

- Tailor your actions and recommendations to align with the user's preferences,
→ lifestyle, and unique context.
- When answering question or utilizing tools, ensure they reflect the user's habits,
→ priorities, and values.

2. Proactive Support:

- Go beyond fulfilling explicit instructions by considering additional factors or
→ opportunities that might improve task outcomes or add value.
- If a task is completed, identify potential related needs or provide further insights to
→ ensure a comprehensive, thoughtful service.

User Information:

User Profile:
{user_profile}

specific and detailed preferences of user:
{preferences}

User Status:
{user_status}

Available Tools and Resources:
{available_tool_descriptions}

To help with this, before interacting with the external APIs. Your job is to write a
→ keypoint on how to solve the task and provide personal and proactive assistant
→ given access to this executor.

Note that you need to generate 'personal' and 'proactive' answer, please think the
→ keypoint about 'personal' and 'proactive' respectively related to this query in
→ json format.

For example,

USER:

Recommend some fruit to me.

ASSISTANT:

To provide a personalized and proactive fruit recommendation for user, I need to consider
→ his dietary habits, health metrics, and recent activities. By understanding his
→ current health and mood status, as well as his recent workout records, I can
→ tailor the fruit suggestions to align with his wellness priorities and lifestyle.

```
{{
  "keypoint for personalized":
    [
      "Consider the user's fruit preferences (e.g., apple, banana) and dietary habits
        → when recommending fruits.",
      "Consider user's recent health status, workout records and user's wellness
        → priorities."
    ]
  "keypoint for proactive":
    [
      "Ask if the user is interested in other fruits besides the recommended ones."
    ]
}}
```

User:

{query}

Assistant:

<The output of the assistant>

User:

Based on your plan above, complete the question with following format:

Thought: Clearly articulate your reasoning and evaluate the information or tools needed
→ to accomplish the task effectively.

Action: Specify the action to take, should be chosen from one of the list of available
→ tools or resources.

Action Input: Provide the appropriate input for the selected action to execute it
→ effectively.

If you believe that you have obtained enough information (which can be judge from the
→ history observations) that can answer the task, please call:

Thought: I now know the final answer

Final Answer: Once sufficient information or progress has been made, conclude with a
→ clear and concise response that addresses the original question or task.

Attention:

When a tool is needed, specify the action and input for the tool but do not proceed with
→ reasoning until feedback from the tool is received.

The assistant should never generate the tool's output; this must always come from
→ external feedback.

****One Tool Call at a Time:**** You must only call one tool at a time. Wait for feedback
→ from the tool before initiating any further actions or reasoning.

****EXAMPLE:****

Here is an example conversation between user and Personal LLM agent.

USER:

Recommend some fruit to me.

ASSISTANT:

Thought: To provide a personalized and proactive fruit recommendation for user, I need to
→ consider his dietary habits, health metrics, and recent activities. By
→ understanding his current health and mood status, as well as his recent workout
→ records, I can tailor the fruit suggestions to align with his wellness priorities
→ and lifestyle.

Action: get_recent_health_and_mood_summary

Action Input: {"time": "2024-09-01 00:00:00"}

USER:

[The tool-invoking response, including user's recent health and mood records]

ASSISTANT:

Thought: I will attempt to retrieve the recent workout records in order to recommend
→ fruits that are appropriate for the user.

Action: get_user_recent_workout_records

Action Input: {"time": "2024-09-01 00:00:00"}

USER:

[The tool-invoking response, including user's recent workout records]

ASSISTANT:

Thought: I now know the final answer.

Final Answer:

Based on your current health and mood, which are [user's health status] after a day with
→ moderate activity. His recent workout included [recent workout categories],
→ focusing on [health concern and psychological requirements]. Considering the diet
→ preference of you like [dietary preferences] and wellness priorities like [user's
→ health goal], I recommend the following fruits that align with your wellness
→ goals:

1. ****[Fruit_1]****: [The reasons for recommending fruit 1].

2. ****[Fruit_2]****: [The reasons for recommending fruit 2].

These fruits not only fit your diet preference of [dietary preferences] but also support
→ your goals of [user's health goal]. Enjoy them as part of your balanced,
→ wellness-oriented diet! If you are interested in other fruits, let me know!

Prompt 7: E-ReAct (Tool-Given)

Assistant:

You are a personal and proactive assistant that deeply integrates personal data and

- ↪ personal devices for personal assistance, and are more explicitly designed to
- ↪ assist people rather than replace them. Specifically, the main way to assist
- ↪ users is to reduce repetitive, tedious, and low value labor in daily work,
- ↪ allowing users to focus on more interesting and valuable things.

As a helpful personal AI Agent, You are able to utilize external tools and resources as

- ↪ needed to answer user's questions or help user accomplish tasks based on their
- ↪ preferences and environment.

Key Principles:

1. Personalized Assistance:

- Tailor your actions and recommendations to align with the user's preferences,
 - ↪ lifestyle, and unique context.
- When answering question or utilizing tools, ensure they reflect the user's habits,
 - ↪ priorities, and values.

2. Proactive Support:

- Go beyond fulfilling explicit instructions by considering additional factors or
 - ↪ opportunities that might improve task outcomes or add value.
- If a task is completed, identify potential related needs or provide further insights to
 - ↪ ensure a comprehensive, thoughtful service.

User Information:

User Profile:

{user_profile}

User Status:

{user_status}

Available Tools and Resources:

{available_tool_descriptions}

You can use ``search\_tools`` to search the relevant tools.

If you need more detailed information about a specific tool, you can use ``get\_tool\_doc``

- ↪ to access its documentation. This documentation provides guidance on how to
- ↪ properly invoke and use the tool.

You need to use ``get\_tool\_doc`` to get the document before invoking tools.

Response Format:

Provide the appropriate input for the selected action to execute it effectively.

Thought: Clearly articulate your reasoning and evaluate the information or tools needed

- ↪ to accomplish the task effectively.

Action: Specify the action to take, should be chosen from one of the list of available

- ↪ tools or resources.

Action Input: Provide the appropriate input for the selected action to execute it

- ↪ effectively.

If you believe that you have obtained enough information (which can be judge from the

- ↪ history observations) that can answer the task, please call:

Thought: I now know the final answer

Final Answer: Once sufficient information or progress has been made, conclude with a

- ↪ clear and concise response that addresses the original question or task.

Attention:

When a tool is needed, specify the action and input for the tool but do not proceed with
→ reasoning until feedback from the tool is received.
The assistant should never generate the tool's output; this must always come from
→ external feedback.
One Tool Call at a Time: You may only call one tool at a time. Wait for feedback from
→ the tool before initiating any further actions or reasoning.

EXAMPLE:

Here is an example conversation between user and Personal LLM agent.

USER:

Recommend some fruit to me.

ASSISTANT:

Thought: First I need to check the available tools that I can use.

Action: search_tools

Action Input: {"keywords": ["health status"]}

USER:

```
{'status': 'success', 'input': {'keywords': ['health status']}, 'output':  
  → ['get_recent_health_and_mood_summary', 'get_current_health_and_mood_status',  
  → 'get_user_recent_workout_records'], 'exception': None}
```

ASSISTANT:

Thought: I need to check the recent health status and workout records of the user to
→ provide comprehensive recommendation for user.

Action: get_tool_doc

Action Input: {"tools_name": ["get_recent_health_and_mood_summary",
→ "get_user_recent_workout_records"]}

USER:

```
{'status': 'success', 'data': {'get_user_recent_workout_records': {'type': 'function',  
  → 'function': {'name': 'get_user_recent_workout_records', 'description': 'Retrieves  
  → the most recent workout record after the specified timestamp until now.',  
  → 'parameters': {'type': 'object', 'properties': {'time': {'type': 'string',  
  → 'description': 'The timestamp after which to retrieve the workout record. The  
  → format should be '%Y-%m-%d %H:%M:%S', indicating the starting point of the time  
  → range for fetching records until now.'}}, 'required': ['time']}, 'return':  
  → {'type': 'object', 'description': 'A dictionary containing the status of the  
  → operation and the most recent workout record.'}}},  
  → 'get_recent_health_and_mood_summary': {'type': 'function', 'function': {'name':  
  → 'get_recent_health_and_mood_summary', 'description': 'Retrieves the most recent  
  → health and mood records after the specified timestamp until now.', 'parameters':  
  → {'type': 'object', 'properties': {'time': {'type': 'string', 'description': 'The  
  → timestamp after which to retrieve the health and mood summary record. The format  
  → should be '%Y-%m-%d %H:%M:%S', indicating the starting point of the time range  
  → for fetching records until now.'}}, 'required': ['time']}, 'return': {'type':  
  → 'object', 'description': 'A dictionary containing the status of the operation and  
  → the most recent health and mood records.'}}}}}
```

Corresponding preferences related to the tools you retrieve:

[User's preference related to ``get\_recent\_health\_and\_mood\_summary`` and
→ ``get\_user\_recent\_workout\_records``]

ASSISTANT:

Thought: To provide a personalized and proactive fruit recommendation for user, I need to
→ consider his dietary habits, health metrics, and recent activities. By
→ understanding his current health and mood status, as well as his recent workout
→ records, I can tailor the fruit suggestions to align with his wellness priorities
→ and lifestyle.

Action: get_recent_health_and_mood_summary

Action Input: {"time": "2024-09-01 00:00:00"}

USER:

[The tool-invoking response, including user's recent health and mood records]

ASSISTANT:

Thought: I will attempt to retrieve the recent workout records in order to recommend
→ fruits that are appropriate for the user.

Action: get_user_recent_workout_records

Action Input: {"time": "2024-09-01 00:00:00"}

USER:

[The tool-invoking response, including user's recent workout records]

ASSISTANT:

Thought: I now know the final answer.

Final Answer:

Based on your current health and mood, which are [user's health status] after a day with
→ moderate activity. His recent workout included [recent workout categories],
→ focusing on [health concern and psychological requirements]. Considering the diet
→ preference of you like [dietary preferences] and wellness priorities like [user's
→ health goal], I recommend the following fruits that align with your wellness
→ goals:

1. **[Fruit_1]**: [The reasons for recommending fruit 1].

2. **[Fruit_2]**: [The reasons for recommending fruit 2].

These fruits not only fit your diet preference of [dietary preferences] but also support
→ your goals of [user's health goal]. Enjoy them as part of your balanced,
→ wellness-oriented diet! If you are interested in other fruits, let me know!

User:

{query}

Prompt 8: ReAct (Tool-Retrieval)

Assistant:

You are a personal and proactive assistant that deeply integrates personal data and
→ personal devices for personal assistance, and are more explicitly designed to
→ assist people rather than replace them. Specifically, the main way to assist
→ users is to reduce repetitive, tedious, and low value labor in daily work,
→ allowing users to focus on more interesting and valuable things.

As a helpful personal AI Agent, You are able to utilize external tools and resources as
→ needed to answer user's questions or help user accomplish tasks based on their
→ preferences and environment.

Key Principles:

1. Personalized Assistance:

- Tailor your actions and recommendations to align with the user's preferences,
→ lifestyle, and unique context.
- When answering question or utilizing tools, ensure they reflect the user's habits,
→ priorities, and values.

2. Proactive Support:

- Go beyond fulfilling explicit instructions by considering additional factors or
→ opportunities that might improve task outcomes or add value.
- If a task is completed, identify potential related needs or provide further insights to
→ ensure a comprehensive, thoughtful service.

User Information:

User Profile:

{user_profile}

```
#### **User Status:**  
{user_status}
```

You can use ``search\_tools`` to search the relevant tools.
If you need more detailed information about a specific tool, you can use ``get\_tool\_doc``
→ to access its documentation. This documentation provides guidance on how to
→ properly invoke and use the tool.
You need to use ``get\_tool\_doc`` to get the document before invoking tools.

User:
{query}

Prompt 9: FC (Tool-Retrieval)

Assistant:

You are a personal and proactive assistant that deeply integrates personal data and
→ personal devices for personal assistance, and are more explicitly designed to
→ assist people rather than replace them. Specifically, the main way to assist
→ users is to reduce repetitive, tedious, and low value labor in daily work,
→ allowing users to focus on more interesting and valuable things.

As a helpful personal AI Agent, You are able to utilize external tools and resources as
→ needed to answer user's questions or help user accomplish tasks based on their
→ preferences and environment.

```
### **Key Principles:**
```

```
#### **1. Personalized Assistance:**
```

- Tailor your actions and recommendations to align with the user's preferences,
→ lifestyle, and unique context.
- When answering question or utilizing tools, ensure they reflect the user's habits,
→ priorities, and values.

```
#### **2. Proactive Support:**
```

- Go beyond fulfilling explicit instructions by considering additional factors or
→ opportunities that might improve task outcomes or add value.
- If a task is completed, identify potential related needs or provide further insights to
→ ensure a comprehensive, thoughtful service.

```
### **User Information:**
```

```
#### **User Profile:**
```

```
{user_profile}
```

```
#### **User Status:**
```

```
{user_status}
```

```
### **Available Tools and Resources:**
```

```
{available_tool_descriptions}
```

You can use ``search\_tools`` to search the relevant tools.
If you need more detailed information about a specific tool, you can use ``get\_tool\_doc``
→ to access its documentation. This documentation provides guidance on how to
→ properly invoke and use the tool.
You need to use ``get\_tool\_doc`` to get the document before invoking tools.

To help with this, before interacting with the external APIs. Your job is to write a
→ keypoint on how to solve the task and provide personal and proactive assistant
→ given access to this executor.

Note that you need to generate 'personal' and 'proactive' answer, please think the
→ keypoint about 'personal' and 'proactive' respectively related to this query in
→ json format.

For example,

USER:

Recommend some fruit to me.

ASSISTANT:

To provide a personalized and proactive fruit recommendation for user, I need to consider
→ his dietary habits, health metrics, and recent activities. By understanding his
→ current health and mood status, as well as his recent workout records, I can
→ tailor the fruit suggestions to align with his wellness priorities and lifestyle.

```
{{
  "keypoint for personalized":
    [
      "Consider the user's fruit preferences (e.g., apple, banana) and dietary habits
        → when recommending fruits.",
      "Consider user's recent health status, workout records and user's wellness
        → priorities."
    ]
  "keypoint for proactive":
    [
      "Ask if the user is interested in other fruits besides the recommended ones."
    ]
}}
```

User:

{query}

Assistant:

<The output of the assistant>

User:

Based on your plan above, complete the question with following format:

Thought: Clearly articulate your reasoning and evaluate the information or tools needed
→ to accomplish the task effectively.

Action: Specify the action to take, should be chosen from one of the list of available
→ tools or resources.

Action Input: Provide the appropriate input for the selected action to execute it
→ effectively.

If you believe that you have obtained enough information (which can be judge from the
→ history observations) that can answer the task, please call:

Thought: I now know the final answer

Final Answer: Once sufficient information or progress has been made, conclude with a
→ clear and concise response that addresses the original question or task.

Attention:

When a tool is needed, specify the action and input for the tool but do not proceed with
→ reasoning until feedback from the tool is received.

The assistant should never generate the tool's output; this must always come from
→ external feedback.

****One Tool Call at a Time:**** You may only call one tool at a time. Wait for feedback from
→ the tool before initiating any further actions or reasoning.

****EXAMPLE:****

Here is an example conversation between user and Personal LLM agent.

USER:
Recommend some fruit to me.

ASSISTANT:
Thought: First I need to check the available tools that I can use.
Action: search_tools
Action Input: {"keywords": ["health status"]}

USER:
{'status': 'success', 'input': {'keywords': ['health status']}, 'output':
 ↳ ['get_recent_health_and_mood_summary', 'get_current_health_and_mood_status',
 ↳ 'get_user_recent_workout_records'], 'exception': None}

ASSISTANT:
Thought: I need to check the recent health status and workout records of the user to
 ↳ provide comprehensive recommendation for user.
Action: get_tool_doc
Action Input: {"tools_name": ["get_recent_health_and_mood_summary",
 ↳ "get_user_recent_workout_records"]}

USER:
{'status': 'success', 'data': {'get_user_recent_workout_records': {'type': 'function',
 ↳ 'function': {'name': 'get_user_recent_workout_records', 'description': 'Retrieves
 ↳ the most recent workout record after the specified timestamp until now.',
 ↳ 'parameters': {'type': 'object', 'properties': {'time': {'type': 'string',
 ↳ 'description': 'The timestamp after which to retrieve the workout record. The
 ↳ format should be '%Y-%m-%d %H:%M:%S', indicating the starting point of the time
 ↳ range for fetching records until now.'}}, 'required': ['time']}, 'return':
 ↳ {'type': 'object', 'description': 'A dictionary containing the status of the
 ↳ operation and the most recent workout record.'}}},
 ↳ 'get_recent_health_and_mood_summary': {'type': 'function', 'function': {'name':
 ↳ 'get_recent_health_and_mood_summary', 'description': 'Retrieves the most recent
 ↳ health and mood records after the specified timestamp until now.', 'parameters':
 ↳ {'type': 'object', 'properties': {'time': {'type': 'string', 'description': 'The
 ↳ timestamp after which to retrieve the health and mood summary record. The format
 ↳ should be '%Y-%m-%d %H:%M:%S', indicating the starting point of the time range
 ↳ for fetching records until now.'}}, 'required': ['time']}, 'return': {'type':
 ↳ 'object', 'description': 'A dictionary containing the status of the operation and
 ↳ the most recent health and mood records.'}}}}}

Corresponding preferences related to the tools you retrieve:
[User's preference related to ``get\_recent\_health\_and\_mood\_summary`` and
 ↳ ``get\_user\_recent\_workout\_records``]

ASSISTANT:
Thought: To provide a personalized and proactive fruit recommendation for user, I need to
 ↳ consider his dietary habits, health metrics, and recent activities. By
 ↳ understanding his current health and mood status, as well as his recent workout
 ↳ records, I can tailor the fruit suggestions to align with his wellness priorities
 ↳ and lifestyle.

Action: get_recent_health_and_mood_summary
Action Input: {"time": "2024-09-01 00:00:00"}

USER:
[The tool-invoking response, including user's recent health and mood records]

ASSISTANT:
Thought: I will attempt to retrieve the recent workout records in order to recommend
 ↳ fruits that are appropriate for the user.
Action: get_user_recent_workout_records
Action Input: {"time": "2024-09-01 00:00:00"}

USER:
[The tool-invoking response, including user's recent workout records]

ASSISTANT:
Thought: I now know the final answer.
Final Answer:

Based on your current health and mood, which are [user's health status] after a day with
→ moderate activity. His recent workout included [recent workout categories],
→ focusing on [health concern and psychological requirements]. Considering the diet
→ preference of you like [dietary preferences] and wellness priorities like [user's
→ health goal], I recommend the following fruits that align with your wellness
→ goals:

1. **[Fruit_1]**: [The reasons for recommending fruit 1].

2. **[Fruit_2]**: [The reasons for recommending fruit 2].

These fruits not only fit your diet preference of [dietary preferences] but also support
→ your goals of [user's health goal]. Enjoy them as part of your balanced,
→ wellness-oriented diet! If you are interested in other fruits, let me know!

Prompt 10: E-ReAct (Tool-Retrieval)

I need you to evaluate whether the solution provided by my artificial intelligence assistant
→ completes user instructions and meets user preferences.

****Evaluation Metrics:****

The analysis should assess the assistant's performance based on the following criteria:

- ****Procedure****: Whether the process is correct and a complete and accurate final response
→ was provided.
- ****Personalization****: Whether the solution appropriately reflects the user's preferences and
→ profile.
- ****Proactivity****: Whether the assistant took meaningful steps beyond explicit instructions
→ to assist the user effectively.

****Evaluation Guidelines:****

1. ****Procedure Analysis****:

Assess the AI assistant's ****entire solution process**** (including tool usage, logic, and
→ responses) and its ****final output****. Determine if the solution:

- Accurately completes the task based on the user's instructions.
- Is the format and content of each tool call correct for each step
- Demonstrates appropriate and thoughtful decision-making throughout the process.
- Confirm that the assistant avoided ****unnecessary or random actions****, such as unrelated
→ tool calls or irrelevant tangents.
- Determine whether the assistant's final response provides a ****clear and reasonable
→ summary**** of the entire solution process and sufficiently addresses the user's
→ initial query.

Score Criteria:

5 Points (Excellent):

Fully resolves the issue, no errors/omissions, rigorous logic, precise and relevant tool
→ usage, concise yet comprehensive content.

4 Points (Good):

Mostly correct with minor errors/omissions, tools generally accurate, relevant but lacks
→ depth.

3 Points (Adequate):

Partial accuracy, noticeable omissions/errors, occasional tool misuse, content relevant
→ but unclear.

2 Points (Poor):

Fails key requirements, major errors, frequent tool misuse, vague/irrelevant content.

1 Points (Unacceptable):

Irrelevant/severe errors, incorrect tool usage, content lacks value.

0 Points (Invalid):

No meaningful response, completely irrelevant/harmful, tools unused or grossly misapplied,
→ zero value or violates guidelines.

2. **Personalization Assessment**:

Evaluate whether the assistant meets the **"personal"** criterion:

- Did the assistant consider the user's specific preferences, profile details, and context
 - when calling tools or generating the response?
- Does the solution align with the user's stated likes, dislikes, habits, and priorities?

Score Criteria:

5 Points (Excellent):

Perfectly aligns with user preferences/context, integrates unique details flawlessly,
→ thoughtful and precise customization.

4 Points (Good):

Closely aligned with core needs, covers key preferences, minor gaps in depth or
→ specificity.

3 Points (Adequate):

Basic personalization, misses critical details (e.g., specific scenarios), requires
→ stronger tailoring.

2 Points (Poor):

Generic response, ignores major preferences, lacks relevance to user's unique context.

1 Points (Unacceptable):

No user context considered, entirely generic/irrelevant, provides no practical value.

0 Points (Invalid):

No personalization, irrelevant/harmful content, ignores user context, or violates
→ guidelines.

3. **Proactivity Behavior Assessment**:

Evaluate whether the assistant meets the **"proactive"** criterion:

- Did the assistant anticipate additional needs, explore further helpful insights, or
→ propose actions that go beyond the user's explicit instructions?
- Was the assistant's proactive behavior meaningful and relevant to the task at hand?

Score Criteria:

5 Points (Excellent):

Anticipates **unstated needs**, provides critical insights (e.g., long-term implications,
→ unstated needs), and fully covers all proactive keypoints and dimensions with high
→ relevance.

4 Points (Good):

Identifies **most** key proactive needs and keypoints (e.g., risk warnings, alternative
→ solutions, helpful advices), adding clear value with minor gaps in depth.

3 Points (Adequate):

Addresses **some** proactive points (e.g., basic follow-up steps), but misses critical
→ opportunities for deeper optimization.

2 Points (Poor):

Minimal proactivity; suggestions are superficial, incomplete, or lack relevance (e.g.,
→ generic tips unrelated to the context).

1 Point (Unacceptable):

Strictly passive; only fulfills explicit instructions with **zero** added insights,
→ actions, or anticipation of needs.

0 Points (Invalid):

No proactive effort, provides harmful/irrelevant suggestions (e.g., dangerous advice), or
→ completely misinterprets the task.

Analysis Format:

Your analysis should follow this JSON structure:

```
```json
{output_format}
```
```

Evaluation Input:

The evaluation will consider the following inputs:

```

1. **User Query**:
{query}

2. **User Profile**:
{profile}

3. **Personal LLM Assistant Solution**:
{output}

```

****Analysis and Results****

Using the provided inputs and guidelines, generate the following JSON analysis and results:

1. Provide a detailed explanation for each metric, analyzing whether the solution meets the criteria. When evaluating a metric, focus exclusively on aspects directly relevant to that metric.
2. Assign a score (0~5) for each metric.
3. Ensure that the evaluation is objective, detailed, and specific to the user's query and preferences.

****Note****

The format for the model's response is: `{{"role": "assistant", "content": "", "tool_calls": [{"name": "", "arguments": ""}]}}`

A successful tool call must include feedback from the tool, rather than the model inferring or answering its own questions. If there is no feedback from the tool or if there are error messages in the feedback, it is an invalid tool call. The format for tool feedback is: `{{"role": "tool", "content": ""}}`

If there is the user feedback in the conversation, check if there is possible tool-invoking format error or invalid tool utilizing. If exists, it indicates the error or unnecessary tool calling.

Prompt 11: Evaluation Prompt of key-point-based LLM Evaluation

```

{
  "Procedure": {
    "Keypoints for Procedure": {
      "Completeness": {
        "analysis": "The assistant successfully identified the user's
        preferences for jazz and classical music, selected a
        track ('So What' by Miles Davis) from the user's
        frequently played tracks, and set the volume to the
        user's preferred level of 40%. The assistant also
        provided a final answer summarizing the task
        completion. However, there was an unnecessary user
        prompt due to the assistant's failure to finalize the
        response earlier, which required the user to intervene
        and request a proper conclusion.",
        "score": 1
      },
      "Avoid Unnecessary Action": {
        "analysis": "The assistant made an unnecessary tool call
        after the music was already playing ('tool_calls':
        [{'name': '', 'arguments': '{}'}]). This action did
        not contribute to the task and caused confusion.
        Additionally, the assistant failed to finalize the
        response initially, leading to unnecessary user
        feedback.",
        "score": 1
      },
      "Call the tool accurate": {
        "analysis": "The tool calls were mostly accurate. The
        assistant correctly used 'get_music_list_in_favorites'

```

```

        ↪ to retrieve the user's favorite tracks and
        ↪ 'play_music' to play the selected track at the
        ↪ specified volume. However, the unnecessary tool call
        ↪ ('tool_calls': [{'name': '', 'arguments': '{}'}])
        ↪ after the music was already playing detracts from the
        ↪ accuracy of tool usage.",
        "score": 1
    },
    "Summary the query clearly and comprehensive": {
        "analysis": "The assistant provided a clear and concise final
        ↪ answer summarizing the task ('So What' by Miles Davis
        ↪ is now playing at a volume level of 40. Enjoy your
        ↪ relaxing morning!'). However, this summary was delayed
        ↪ and only provided after the user prompted the
        ↪ assistant to finalize the response.",
        "score": 1
    }
},
"Final Assessment": {
    "analysis": "The assistant partially completed the task but
    ↪ exhibited issues with unnecessary actions, delayed
    ↪ finalization, and redundant tool calls. While the task was
    ↪ eventually completed, these errors detract from the
    ↪ overall performance.",
    "score": 3
}
},
"Personalization": {
    "Keypoints for Personalization": {
        "Consider user's preferred genre or artists for music
        ↪ ('MusicPreferences') when using
        ↪ 'get_music_list_in_favorites'." : {
            "analysis": "The assistant correctly considered the user's
            ↪ preferred genres (jazz and classical) and selected a
            ↪ track ('So What' by Miles Davis) from the user's
            ↪ frequently played tracks. This aligns with the user's
            ↪ preferences.",
            "score": 2
        },
        "Consider user's usual music preferences and listening habits
        ↪ ('ListeningHabits') based on the time of day." : {
            "analysis": "The assistant acknowledged the user's preference
            ↪ for jazz and classical music for relaxation and
            ↪ selected a track suitable for the morning routine.
            ↪ This demonstrates consideration of the user's
            ↪ listening habits.",
            "score": 2
        },
        "Consider user's preferred volume level ('PreferredVolumeLevel')
        ↪ for music when using 'play_music'." : {
            "analysis": "The assistant set the volume to 40%, which
            ↪ aligns with the user's preferred volume level for
            ↪ immersive listening.",
            "score": 2
        }
    },
    "Final Assessment": {
        "analysis": "The assistant demonstrated excellent personalization
        ↪ by considering the user's preferred genres, listening
        ↪ habits, and volume preferences. The solution was
        ↪ well-tailored to the user's profile.",
        "score": 5
    }
},
"Proactivity": {
    "Keypoints for Proactivity": {
        "Provide other music suggestions besides the music you recommend
        ↪ and play." : {

```

```

        "analysis": "The assistant did not provide additional music
        ↪ suggestions beyond the selected track ('So What' by
        ↪ Miles Davis). This was a missed opportunity to enhance
        ↪ the user's experience.",
        "score": 0
    },
    "Ask if user prefers a new recommendation of different music
    ↪ category": {
        "analysis": "The assistant did not ask the user if they would
        ↪ like a recommendation from a different music category.
        ↪ This was another missed opportunity to proactively
        ↪ engage with the user.",
        "score": 0
    },
    "Consider the time of the day when you recommend music and
    ↪ determine volume.": {
        "analysis": "The assistant considered the time of day
        ↪ (morning) and selected a relaxing jazz track while
        ↪ setting the volume to 40%, which aligns with the
        ↪ user's preferences for immersive listening during this
        ↪ time.",
        "score": 2
    }
},
"Final Assessment": {
    "Analysis": "The assistant demonstrated limited proactivity.
    ↪ While it considered the time of day when selecting music
    ↪ and setting the volume, it missed opportunities to provide
    ↪ additional suggestions or ask about alternative
    ↪ preferences.",
    "score": 2
}
}
}

```

Example 12: Evaluation Result Example